Resume screening app using LLMs and NLP.

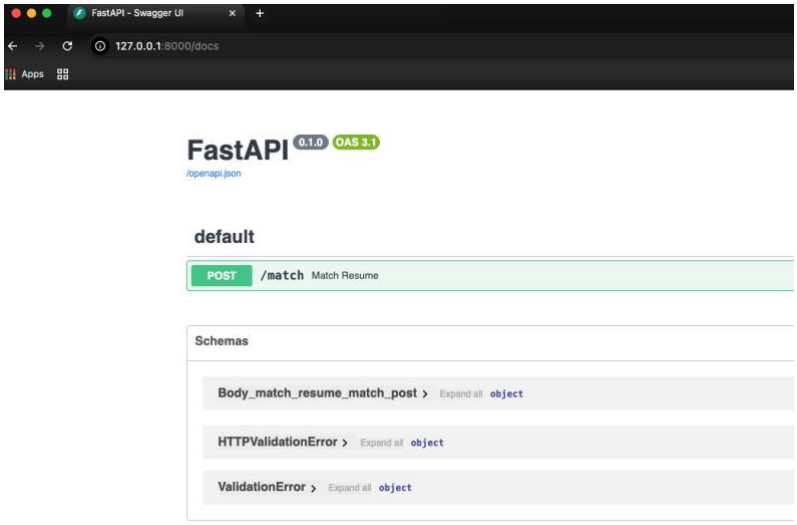
Installed the required tools.

```
sh: command not found: app.py
(base) pnsingh@PNs-MacBook-Air resume-screening_ss % touch app.py
(base) pnsingh@PNs-MacBook-Air resume-screening_ss % touch requirements.txt
(base) pnsingh@PNs-MacBook-Air resume-screening_ss % nano requirements.txt
(base) pnsingh@PNs-MacBook-Air resume-screening_ss % pip install -r requirements.txt
ollecting fastapi (from -r requirements.txt (line 1))
 Downloading fastapi-0.135.1-py3-none-any.whl.metadata (30 kB)
ollecting uvicorn (from -r requirements.txt (line 2))
 Downloading uvicorn-0.42.0-py3-none-any.whl.metadata (6.7 kB)
equirement already satisfied: scikit-learn in /opt/anaconda3/lib/python3.12/site-packages (from -r requirements.txt (line 3)) (1.4.2)
ollecting python-multipart (from -r requirements.txt (line 4))
 Downloading python_multipart-0.0.22-py3-none-any.whl.metadata (1.8 kB)
ollecting PyPDF2 (from -r requirements.txt (line 5))
 Downloading pypdf2-3.0.1-py3-none-any.whl.metadata (6.8 kB)
ollecting starlette>=0.46.0 (from fastapi->-r requirements.txt (line 1))
 Downloading starlette-0.52.1-py3-none-any.whl.metadata (6.3 kB)
ollecting pydantic>=2.7.0 (from fastapi->-r requirements.txt (line 1))
 Downloading pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 90.6/90.6 kB 2.1 MB/s eta 0:00:00
equirement already satisfied: typing-extensions>=4.8.0 in /opt/anaconda3/lib/python3.12/site-packages (from fastapi->-r requirements.txt (line 1)) (4.11.0)
ollecting typing-inspection>=0.4.2 (from fastapi->-r requirements.txt (line 1))
 Downloading typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
ollecting annotated-doc>=0.0.2 (from fastapi->-r requirements.txt (line 1))
 Downloading annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
equirement already satisfied: click>=7.0 in /opt/anaconda3/lib/python3.12/site-packages (from uvicorn->-r requirements.txt (line 2)) (8.1.7)
equirement already satisfied: h11>=0.8 in /opt/anaconda3/lib/python3.12/site-packages (from uvicorn->-r requirements.txt (line 2)) (0.14.0)
equirement already satisfied: numpy>=1.19.5 in /opt/anaconda3/lib/python3.12/site-packages (from scikit-learn->-r requirements.txt (line 3)) (1.26.4)
equirement already satisfied: scipy>=1.6.0 in /opt/anaconda3/lib/python3.12/site-packages (from scikit-learn->-r requirements.txt (line 3)) (1.13.1)
equirement already satisfied: joblib>=1.2.0 in /opt/anaconda3/lib/python3.12/site-packages (from scikit-learn->-r requirements.txt (line 3)) (1.4.2)
equirement already satisfied: threadpoolctl>=2.0.0 in /opt/anaconda3/lib/python3.12/site-packages (from scikit-learn->-r requirements.txt (line 3)) (2.2.0)
equirement already satisfied: annotated-types>=0.6.0 in /opt/anaconda3/lib/python3.12/site-packages (from pydantic>=2.7.0->fastapi->-r requirements.txt (line 1)) (0.6.0)
ollecting pydantic-core==2.41.5 (from pydantic>=2.7.0->fastapi->-r requirements.txt (line 1))
 Downloading pydantic_core-2.41.5-cp312-cp312-macosx_11_0_arm64.whl.metadata (7.3 kB)
ollecting typing-extensions>=4.8.0 (from fastapi->-r requirements.txt (line 1))
 Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
equirement already satisfied: anyio<5,>=3.6.2 in /opt/anaconda3/lib/python3.12/site-packages (from starlette>=0.46.0->fastapi->-r requirements.txt (line 1)) (4.2.0)
equirement already satisfied: idna>=2.8 in /opt/anaconda3/lib/python3.12/site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi->-r requirements.txt (line 1)) (3.7)
equirement already satisfied: sniffio>=1.1 in /opt/anaconda3/lib/python3.12/site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0->fastapi->-r requirements.txt (line 1)) (1.3.0)
ownloading fastapi-0.135.1-py3-none-any.whl (116 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 117.0/117.0 kB 5.7 MB/s eta 0:00:00
ownloading uvicorn-0.42.0-py3-none-any.whl (68 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 68.8/68.8 kB 6.8 MB/s eta 0:00:00
ownloading python_multipart-0.0.22-py3-none-any.whl (24 kB)
ownloading pypdf2-3.0.1-py3-none-any.whl (232 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 232.6/232.6 kB 11.7 MB/s eta 0:00:00
ownloading annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
ownloading pydantic-2.12.5-py3-none-any.whl (463 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 463.7/463.7 kB 17.6 MB/s eta 0:00:00
ownloading pydantic_core-2.41.5-cp312-cp312-macosx_11_0_arm64.whl (1.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.9/1.9 MB 26.5 MB/s eta 0:00:00
ownloading starlette-0.52.1-py3-none-any.whl (74 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 74.3/74.3 kB 8.3 MB/s eta 0:00:00
ownloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.6/44.6 kB 4.5 MB/s eta 0:00:00
ownloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
nstalling collected packages: uvicorn, typing-extensions, python-multipart, PyPDF2, annotated-doc, typing-inspection, starlette, pydantic-core, pydantic, fastapi
 Attempting uninstall: typing-extensions
   Found existing installation: typing_extensions 4.11.0
   Uninstalling typing_extensions-4.11.0:
     Successfully uninstalled typing_extensions-4.11.0
 Attempting uninstall: pydantic-core
   Found existing installation: pydantic_core 2.14.6
   Uninstalling pydantic_core-2.14.6:
     Successfully uninstalled pydantic_core-2.14.6
 Attempting uninstall: pydantic
   Found existing installation: pydantic 2.5.3
   Uninstalling pydantic-2.5.3:
     Successfully uninstalled pydantic-2.5.3
uccessfully installed PyPDF2-3.0.1 annotated-doc-0.0.4 fastapi-0.135.1 pydantic-2.12.5 pydantic-core-2.41.5 python-multipart-0.0.22 starlette-0.52.1 typing-extensions-4.15.0 typing-inspection-0.4.2 uvicorn-0.42.0
(base) pnsingh@PNs-MacBook-Air resume-screening_ss %
```

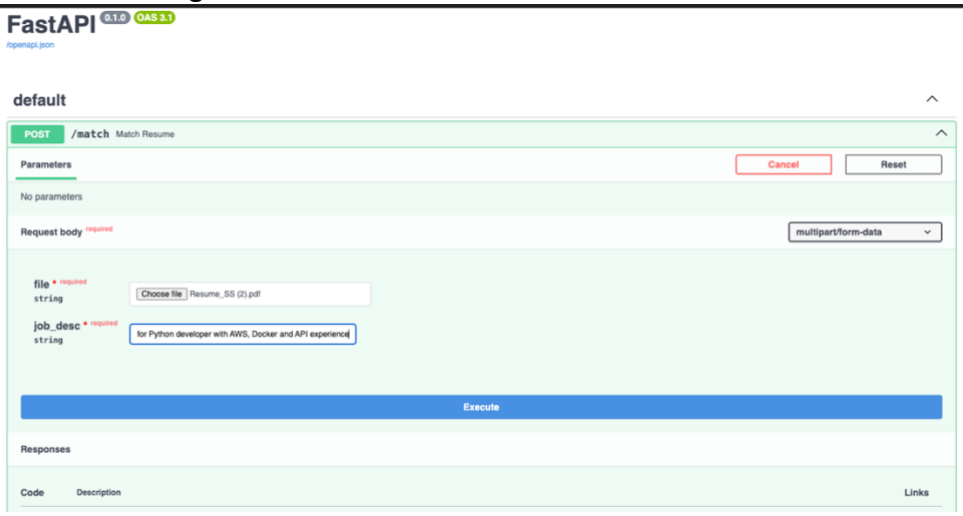
Running uvicorn in terminal.

```
Successfully installed PyPDF2-3.0.1 annotated-doc-0.0.4 fastapi-0.135.1 pydantic-2.12.5 pydantic-core-2.41.5 python-multipart-0.0.22 starlette-0.52.1 typing-extensions-4.15.0 typing-inspection-0.4.2 uvicorn-0.4
(base) pnsingh@PNs-MacBook-Air resume-screening_ss % nano app.py
(base) pnsingh@PNs-MacBook-Air resume-screening_ss % uvicorn app:app --reload
INFO:     Will watch for changes in these directories: ['/Users/pnsingh/resume-screening_ss']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [22393] using StatReload
INFO:     Started server process [22398]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:60023 - "GET /docs HTTP/1.1" 200 OK
INFO:     127.0.0.1:60023 - "GET /openapi.json HTTP/1.1" 200 OK
```

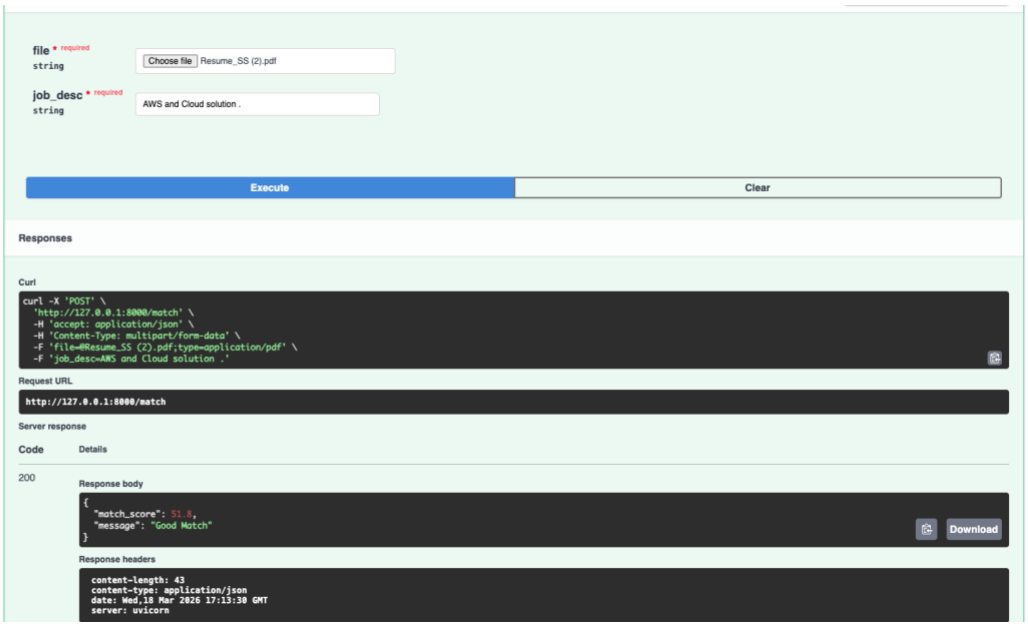
UI Swagger



Tested if the API is Working or NOT.

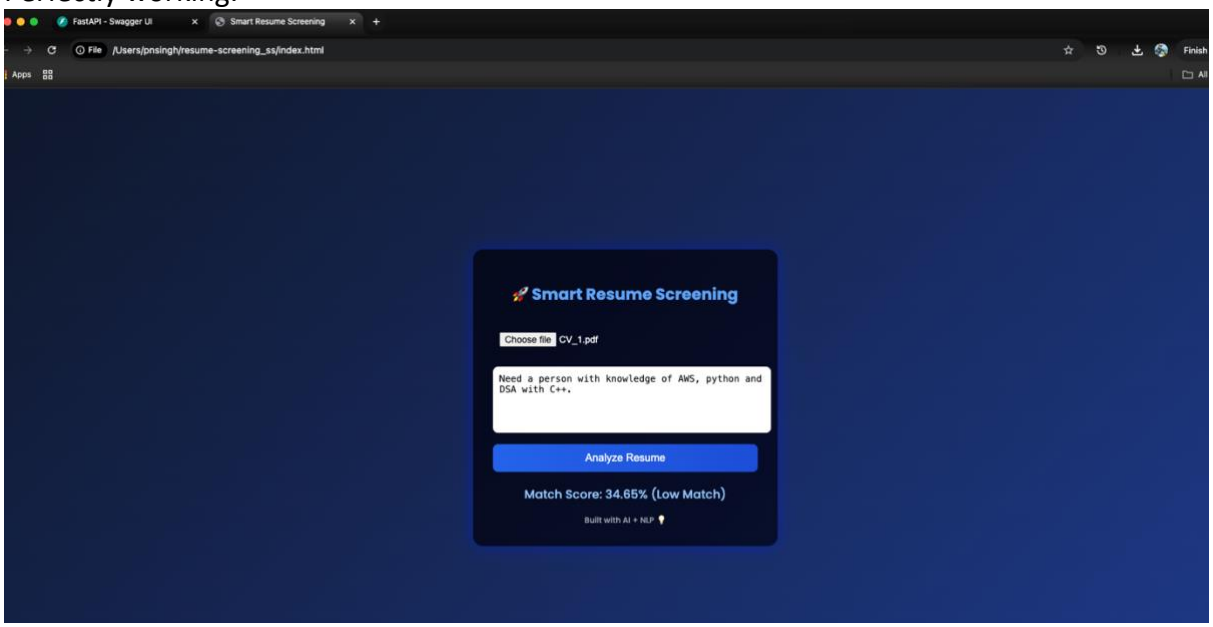


Working Fine.

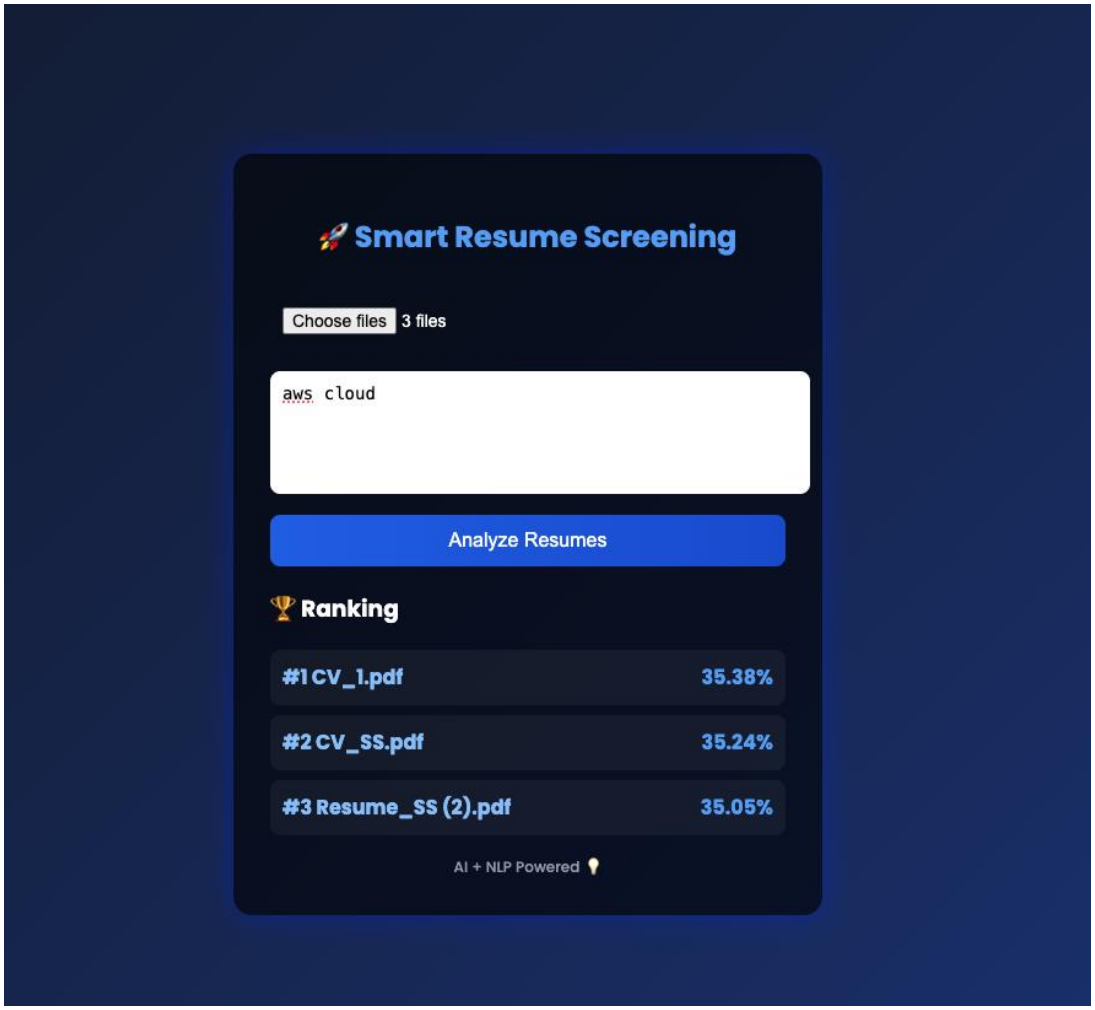


Made Changes in the User Interface to look good.

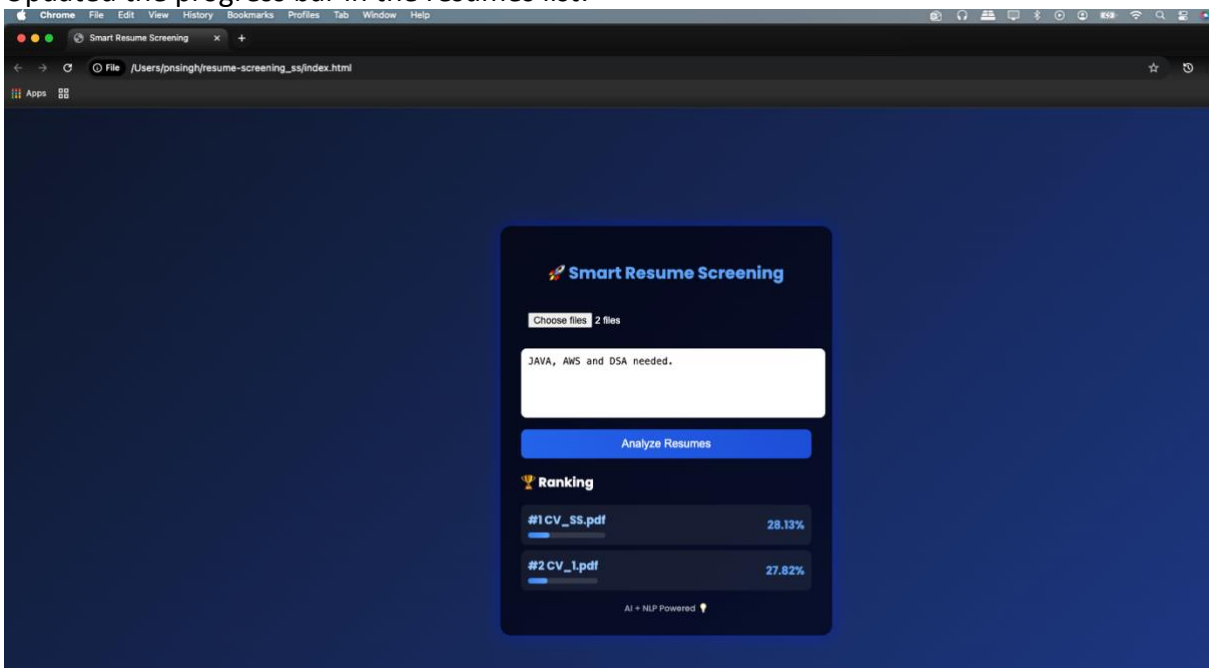
Perfectly working.



Updated the app.py and index.html for taking multiple files at once and giving result.



Updated the progress bar in the resumes list.



Updated the file for skill Extraction in Backend.

```python
def extract_skills(text):
    skills_list = [
        "python", "java", "aws", "docker", "kubernetes",
        "sql", "machine learning", "data analysis",
        "react", "node", "api", "tensorflow", "pandas"
    ]

    text = text.lower()
    found_skills = []

    for skill in skills_list:
        if skill in text:
            found_skills.append(skill)

    return found_skills
```

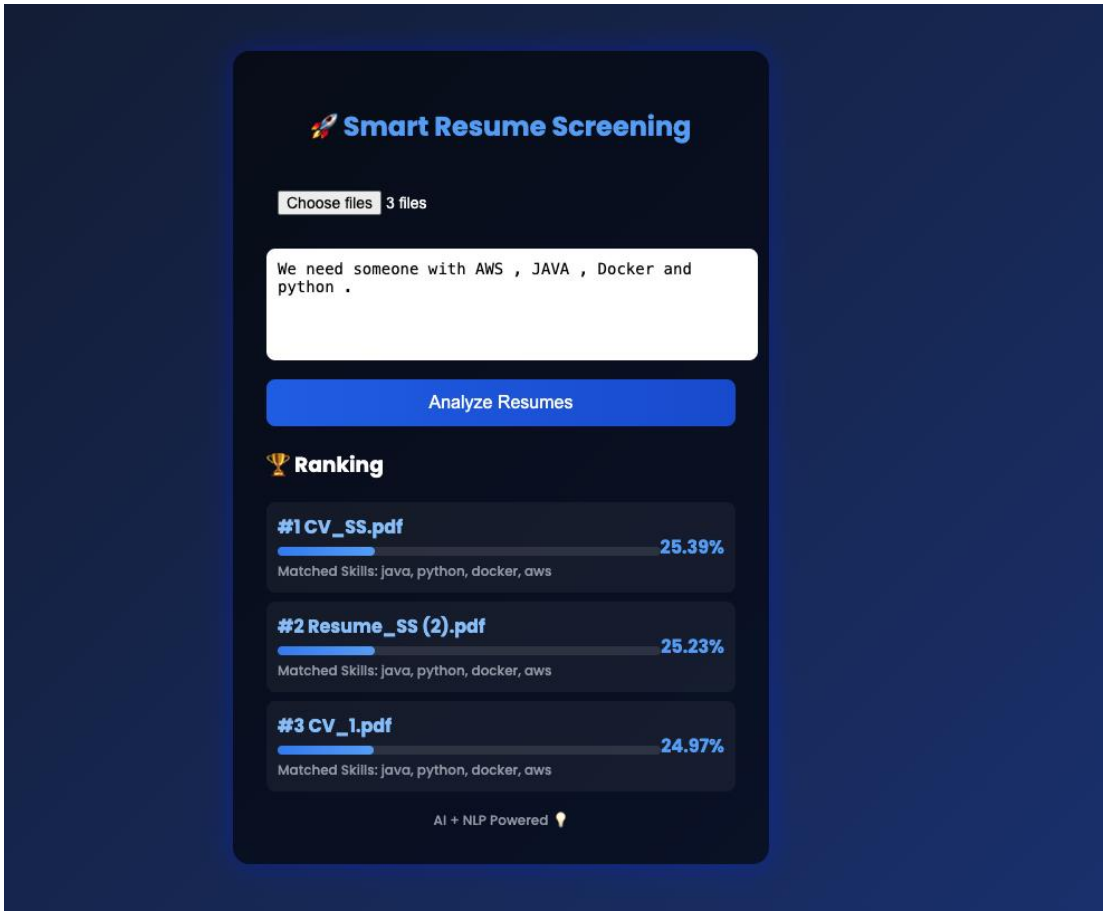
```python
resume_skills = extract_skills(text)
jd_skills = extract_skills(job_desc)

matching_skills = list(set(resume_skills) & set(jd_skills))

results.append({
    "name": filename,
    "score": score,
    "skills": resume_skills,
    "matched": matching_skills
})
```

Now Showing Matched Skills.

My system not only ranks resumes but also explains why a candidate matches by extracting and comparing skills

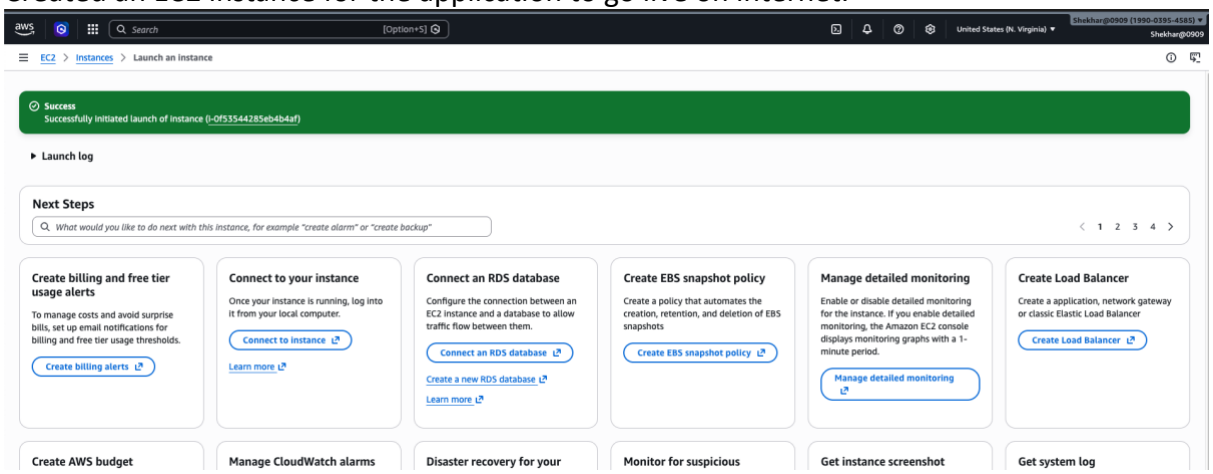


Initialised Connection with docker .Connected with Docker and running.

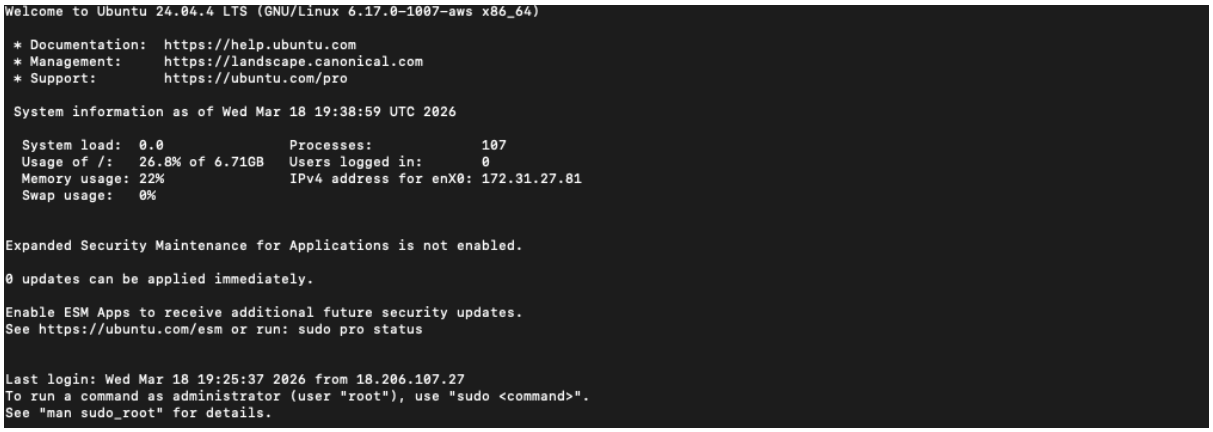
```
 seccomp
  Profile: builtin
 cgroupns
Kernel Version: 6.12.72-linuxkit
Operating System: Docker Desktop
OSType: linux
Architecture: aarch64
CPUs: 8
Total Memory: 3.827GiB
Name: docker-desktop
ID: 5ff4cad0-b160-4527-b2b1-cd42180ad0cc
Docker Root Dir: /var/lib/docker
Debug Mode: false
HTTP Proxy: http.docker.internal:3128
HTTPS Proxy: http.docker.internal:3128
No Proxy: hubproxy.docker.internal
Labels:
 com.docker.desktop.address=unix:///Users/pnsingh/Library/Containers/com.docker.docker/Data/docker-cli.sock
Experimental: false
Insecure Registries:
 hubproxy.docker.internal:5555
 ::1/128
 127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables

base) pnsingh@PNs-MacBook-Air resume-screening_ss % docker build -t resume-app .
+] Building 47.8s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 223B
=> [internal] load metadata for docker.io/library/python:3.10
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 66B
=> [1/5] FROM docker.io/library/python:3.10@sha256:70a693a5ab49ada7d4d5d974678288262bfeccadf06b8362c90ec9cd1a9b7c97
=> => resolve docker.io/library/python:3.10@sha256:70a693a5ab49ada7d4d5d974678288262bfeccadf06b8362c90ec9cd1a9b7c97
=> => sha256:7ca168f25d2959c5e36e85e22b60c1b371a678041338f73ed6a7bb084e7da38a 249B / 249B
=> => sha256:0cbad1ea40b8ff4af34d3d2ee345edae84a03842f6ac92365adfeb1c326de94d 6.19MB / 6.19MB
=> => sha256:f4a64930b7a2ba2c3d82f44eff6485d9753905dcd36f1aafb7f02967cdc6e82a 20.63MB / 20.63MB
=> => sha256:86cdc6c7463a47d1f18421cbc086ad744c8d50c71a79c199d3f739370d14f166 226.21MB / 226.21MB
=> => sha256:d16813bdedcf0a1acbd78336c5bed6fbfaee2674574408140ba7e896cd49cb95 67.58MB / 67.58MB
=> => sha256:6778b62bd97b31237948877e89abc29ad2b2c003f3b965027457c8c90d4f44e0 25.02MB / 25.02MB
=> => sha256:deab8db42772fcfeb45003461fe0eb7bf63bdb29199a4fb76242b9493437c28b 49.66MB / 49.66MB
=> => extracting sha256:deab8db42772fcfeb45003461fe0eb7bf63bdb29199a4fb76242b9493437c28b
=> => extracting sha256:6778b62bd97b31237948877e89abc29ad2b2c003f3b965027457c8c90d4f44e0
=> => extracting sha256:d16813bdedcf0a1acbd78336c5bed6fbfaee2674574408140ba7e896cd49cb95
=> => extracting sha256:86cdc6c7463a47d1f18421cbc086ad744c8d50c71a79c199d3f739370d14f166
=> => extracting sha256:0cbad1ea40b8ff4af34d3d2ee345edae84a03842f6ac92365adfeb1c326de94d
=> => extracting sha256:f4a64930b7a2ba2c3d82f44eff6485d9753905dcd36f1aafb7f02967cdc6e82a
=> => extracting sha256:7ca168f25d2959c5e36e85e22b60c1b371a678041338f73ed6a7bb084e7da38a
=> [internal] load build context
=> => transferring context: 375.46kB
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:8340a9e2d1dec7a5e5028a6ef1b30c98b9faa4bd49d259cbd5ad428ba4b3600b
=> => exporting config sha256:6de29151a1314cd3ed4a635938162a55d87f6712ffca1e8353b2e6ad8c820de7
=> => exporting attestation manifest sha256:7440c8c58b62798992cd9b37599207b56909a8bce5b19c88ddee0c9006775df3
=> => exporting manifest list sha256:81edfd1e7a9176038a185b296366b5567cf8a5c6e4b98043616cfacf8bfe5ed3
=> => naming to docker.io/library/resume-app:latest
=> => unpacking to docker.io/library/resume-app:latest
base) pnsingh@PNs-MacBook-Air resume-screening_ss % docker run -p 8000:8000 resume-app
NFO:     Started server process [1]
NFO:     Waiting for application startup.
NFO:     Application startup complete.
NFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

Created an EC2 instance for the application to go live on internet.



Connected successfully with Ubuntu.



Successfully running On AWS EC2.Project is now live on internet.

```
Successfully built f822d599e7b5
Successfully tagged resume-app:latest
ubuntu@ip-172-31-27-81:~/resume-screening_ss$ sudo docker run -p 8000:8000 resume-app
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     103.47.74.66:6552 - "GET /docs HTTP/1.1" 200 OK
^CINFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [1]
ubuntu@ip-172-31-27-81:~/resume-screening_ss$ sudo docker run -p 8000:8000 resume-app
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

i-0f53544285eb4b4af (resume-app)

PublicIPs: 34.234.65.118 PrivateIPs: 172.31.27.81